

EXPERIMENT NO: 1

Evidence Acquisition Using AccessData FTK Imager

DATE:

1. AIM :

To perform a complete forensic acquisition and analysis lifecycle by imaging non-volatile storage using AccessData FTK Imager

2. DESCRIPTION :

A thorough digital forensic investigation requires collecting both data at rest and data in motion using strict operational procedures.

- **Non-Volatile Memory Acquisition:** AccessData FTK Imager creates mathematically verifiable, bit-by-bit physical images of storage devices (saved as Raw/dd) without altering the original evidence.
- **Volatile Memory (RAM) Acquisition:** Command-line utilities like **WinPmem** load a specialized kernel driver (pme98A2.tmp) to read physical memory directly. Executing this via CMD writes active processes, network connections, and decrypted keys to a raw dump file (.raw).
- **Volatile Memory Analysis (CLI): Volatility 3** is a framework for analyzing RAM dumps. By passing specific parameters and plugins (e.g., -f <image_path> windows.pslist.PsList) into the command line or workbench, investigators can reconstruct the system's live state and identify suspicious processes.

3. PROCEDURE :

Part A: Acquiring Non-Volatile Memory (Disk Image) using FTK Imager

Step 1: Select the Source Type

Open FTK Imager as an Administrator. Navigate to **File > Create Disk Image**. In the Select Source dialog, choose **Physical Drive**.

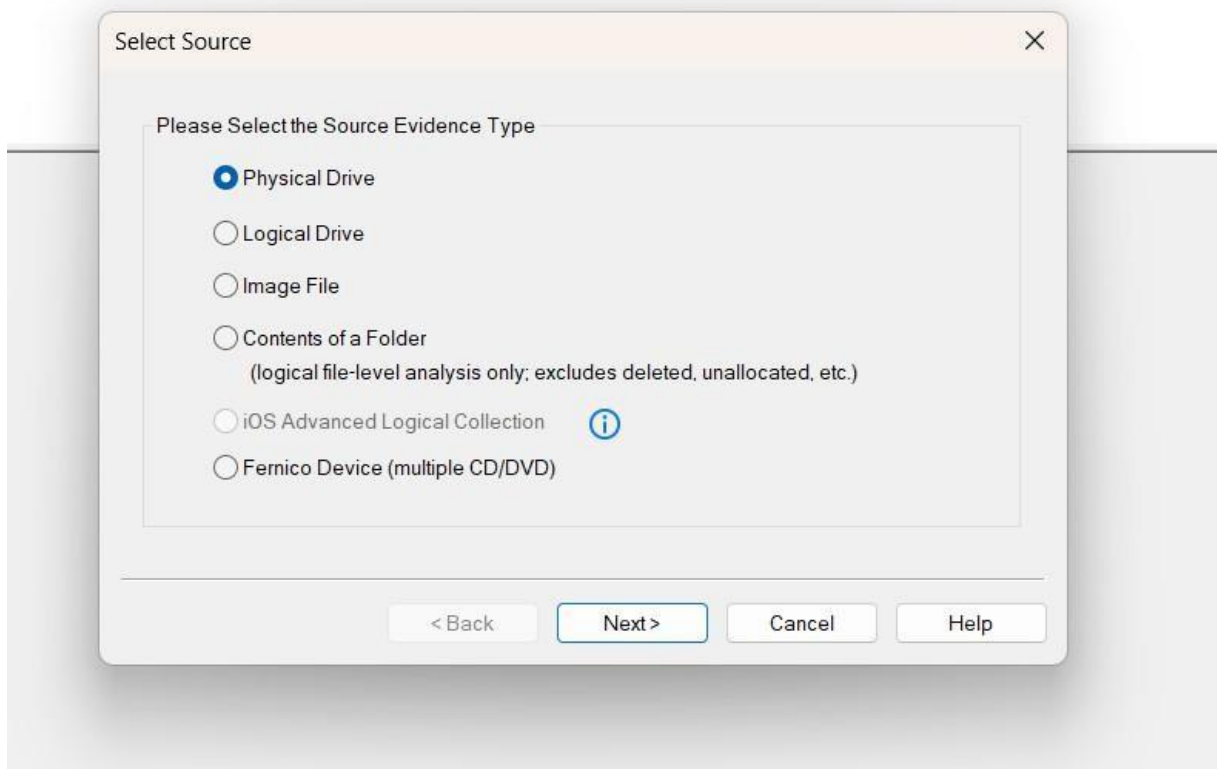


Figure 1: Selecting Physical Drive as the source evidence type

Step 2: Select the Specific Drive

Select the target evidence drive (from the dropdown.

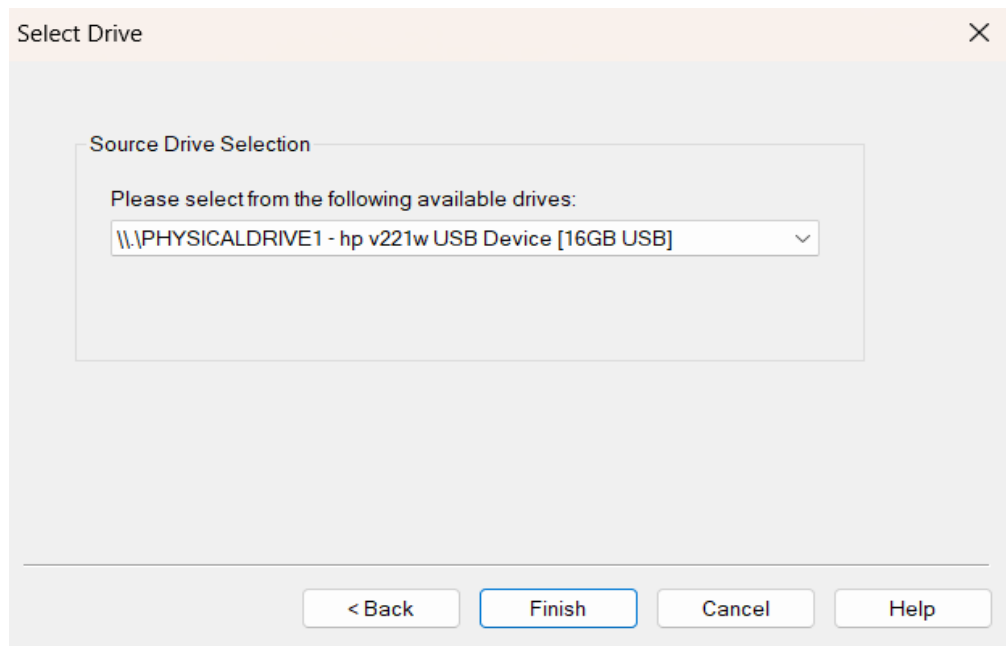


Figure 2: Selecting the target USB device for acquisition

Step 3: Choose the Image Type

Select **Raw (dd)** as the destination image type to create a pure bitstream copy.

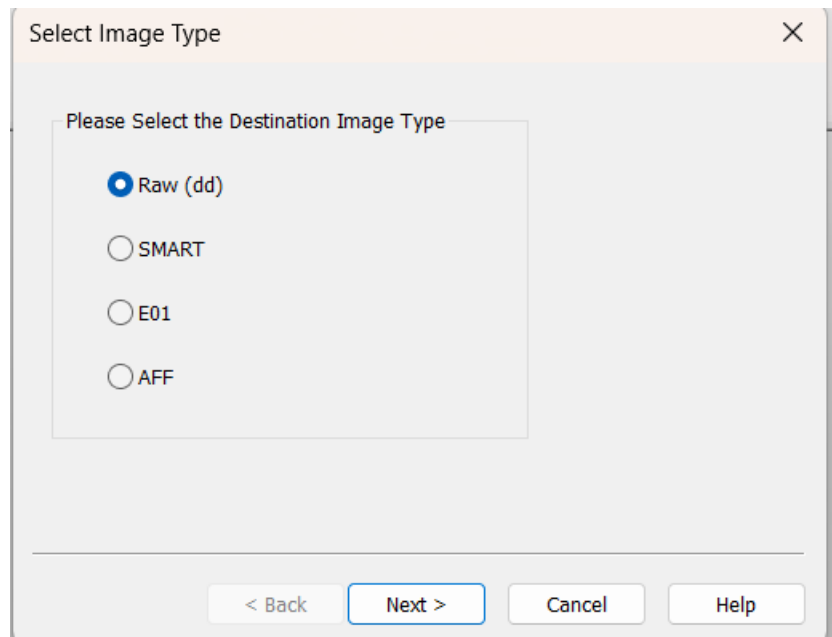


Figure 3: Configuring the destination image format to Raw (dd)

Step 4: Enter Evidence Item Information

Provide the standardized case details to maintain the chain of custody:

- **Case Number:** 99240040210
- **Evidence Number:** 1
- **Unique Description:** Examine
- **Examiner:** sadiq

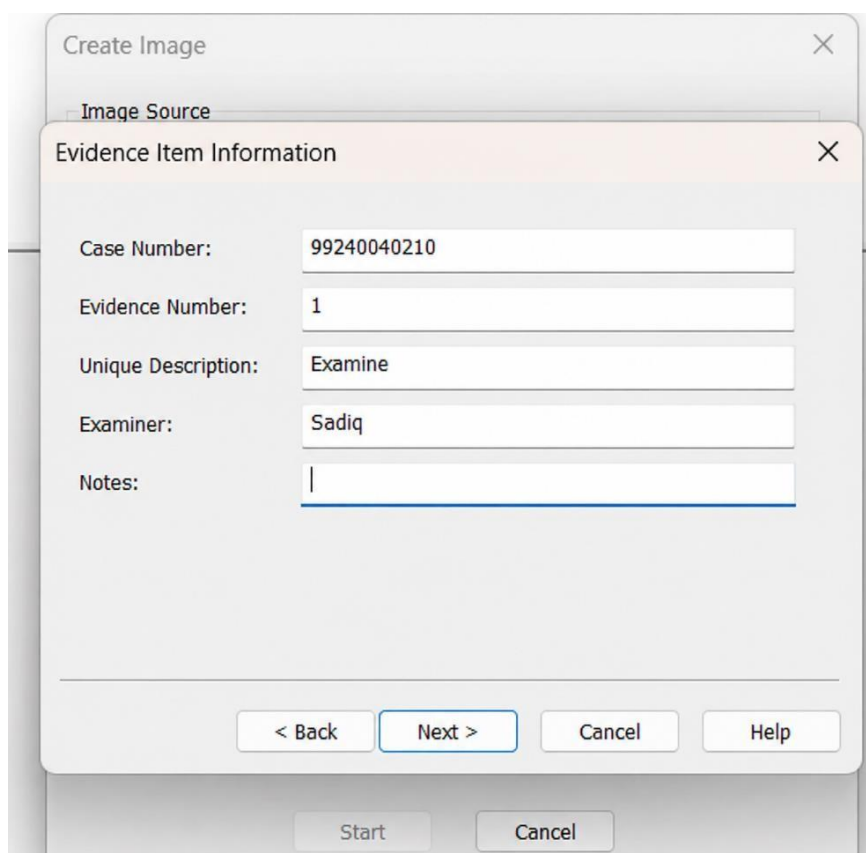


Figure 4: Entering the forensic case and examiner details

Step 5: Configure Image Destination

Set the destination folder and the filename .Set the Fragment Size to 20 MB and check Verify images after they are created.

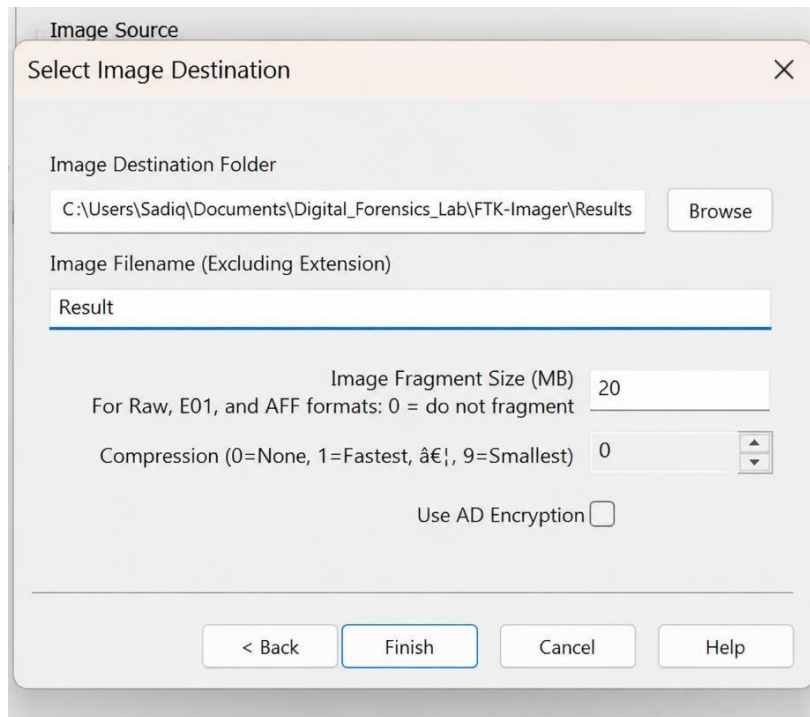


Figure 5: Setting the destination path, filename, and fragmentation size

Step 6: Monitor Acquisition Progress & Hash Verification

Click **Start**. Once complete, FTK Imager hashes the original drive and the newly created image. Review the Drive/Image Verify Results.

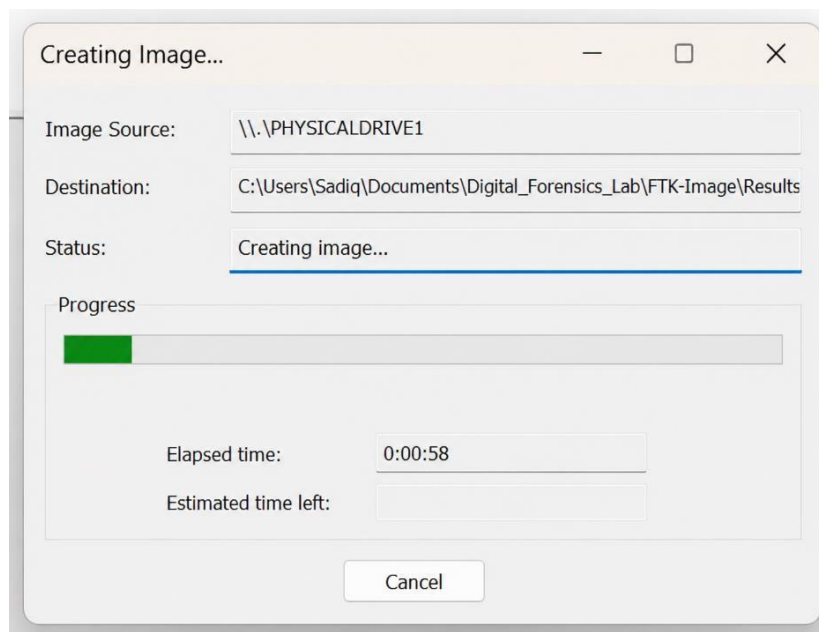


Figure 6: Disk imaging progress tracking

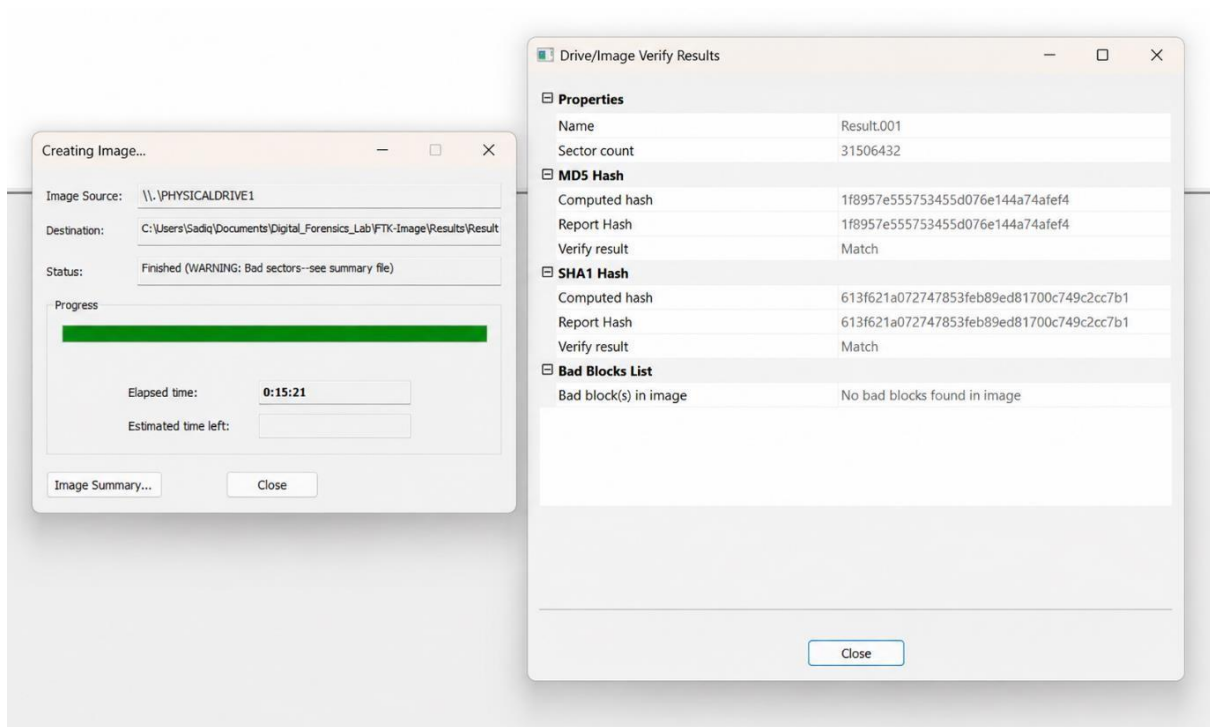


Figure 7: Successful verification of MD5 and SHA1 hashes

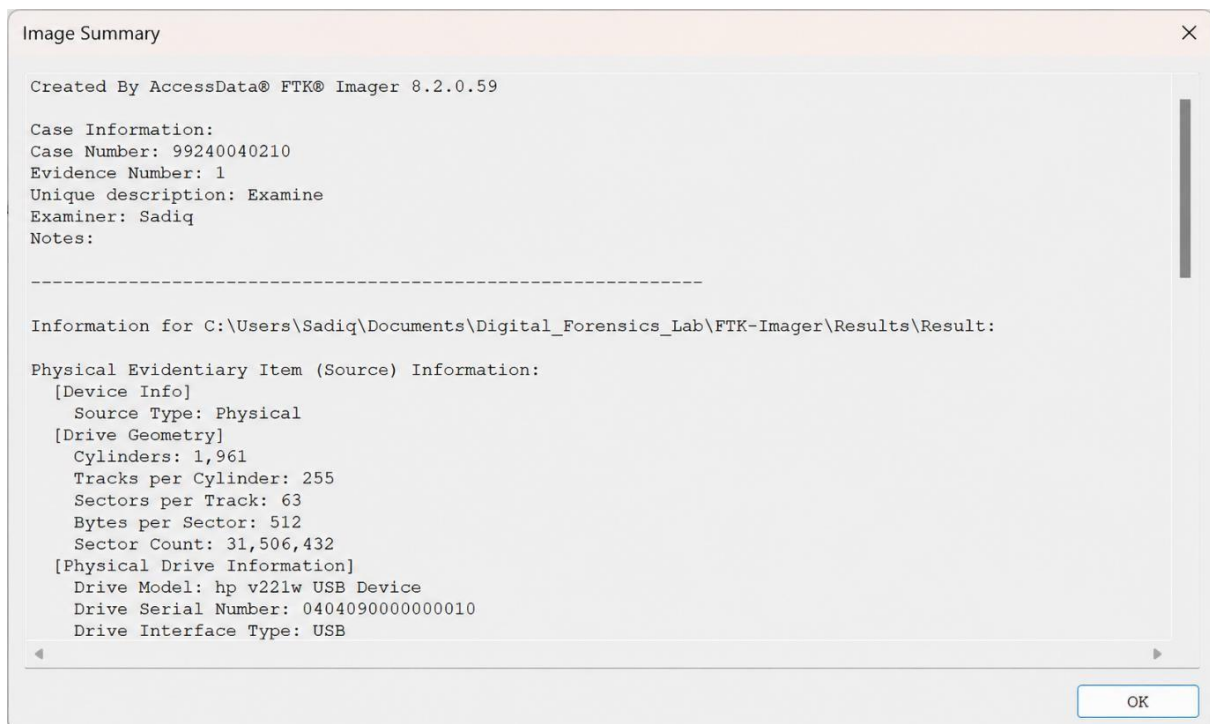


Figure 8: Image Summary detailing the physical device geometry

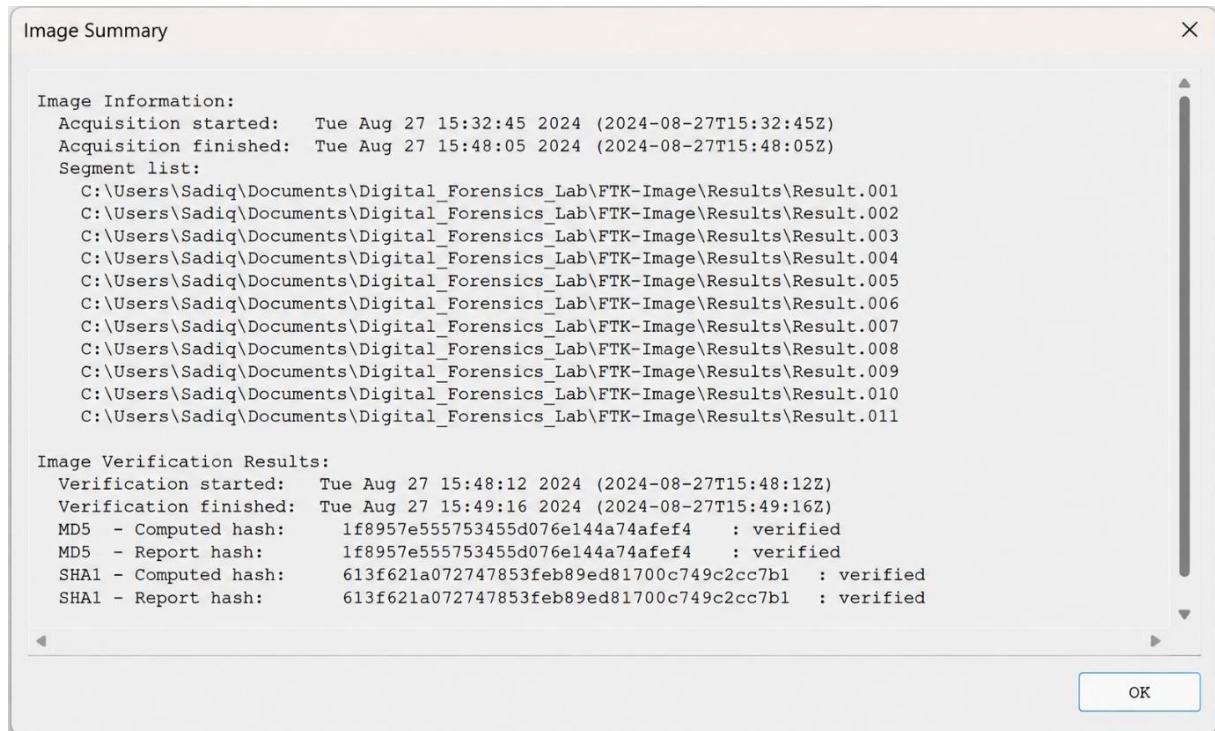


Figure 9: Image Summary detailing the hash verification results and segment list

4. RUBRICS :

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

5. RESULT:

A complete digital forensics acquisition and analysis workflow was successfully executed . The target physical USB drive was securely imaged and verified using FTK Imager. Hash values matched